

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've spent 10 years building backend systems and infrastructure, with the last 3 focused on technical leadership and architecture. I design event-driven, multi-tenant systems on AWS: APIs serving 5 teams with 99.9% uptime processing \$4.3M in transactions, microservices on Kubernetes handling millions of events, and developer platforms that reduced provisioning by 60x. I work in Python, Java, and TypeScript with Postgres, Redis, and Elasticsearch. I've both led teams directly and led through technical vision via RFCs, mentorship, and code quality. I own technical domains from architecture through operations and maintenance.

Work Experience

Alberta Motor Association

Edmonton, Alberta

SOFTWARE ARCHITECT

Jan 2023 - Present

- Built self-service developer platform solo that reduced environment provisioning from 2-3 weeks to 2 hours (60x improvement). Adopted by 15 engineers across 3 teams, created 120+ environments. Unblocked major initiatives by enabling rapid iteration and production debugging. (Python, Terraform, AWS, Jenkins, Docker)
- Own API architecture for platform serving 5 internal teams and external partners. Designed API contracts, approve endpoint additions, maintain OpenAPI standards, and ensure consistency across services. APIs process \$4.3M in annual transactions. (Python, REST, OpenAPI, TypeScript)
- Architected technical infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, \$4.3M annual revenue, 2.8M events processed, 99.9% uptime. Won 2024 Guidewire Innovation Award. (Python, AWS Lambda, Terraform, TypeScript)
- Led modernization of document generation system from legacy C# to containerized microservices on EKS. Led 2-3 engineers. Integrated Datadog for observability (failure rates, success metrics, performance). Reduced environment provisioning from weeks to minutes. (Kubernetes, EKS, Docker, Datadog, Helm, C#)
- Set technical standards and provide architectural guidance for 12-person team through code reviews, RFC authorship, and technical documentation. Member of Architecture Steering Committee. Mentor engineers on system design, debugging, and best practices.

Arterys Inc. (Acquired by Tempus Labs)

Calgary, Alberta (Remote)

SOFTWARE DEVELOPER

Aug 2021 - Aug 2022

- Built backend services running medical AI algorithms, optimizing for low latency and high throughput. (TypeScript, Node.js, AWS)
- Improved algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, identifying and resolving issues before customer impact.

Alberta Motor Association

Edmonton, Alberta

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework from scratch that reduced defects by 30% on subsequent releases. Framework adopted org-wide, still in use 4+ years later across all development teams. Significantly improved developer velocity and release confidence. (Java/Gosu, JUnit)
- Led 5-6 person development team using TDD methodology. Delivered projects on time with no critical defects post-launch. Conducted code reviews, mentored junior developers, and collaborated with stakeholders on requirements and priorities.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local dev environments, reducing onboarding friction for new developers.

Pleasant Solutions

Edmonton, Alberta

SOFTWARE DEVELOPER

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD. (C#)

Lockheed Martin Canada

Halifax, Nova Scotia

SOFTWARE ENGINEER ASSOCIATE

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining real-time data streams from multiple sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system for capturing and replaying test scenarios, improving debugging and test coverage.

The King's University

Edmonton, Alberta

SOFTWARE DEVELOPER/RESEARCH ASSISTANT

May 2014 - Sep 2016

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.

Education

The Open University

MSC. COMPUTING (SOFTWARE ENGINEERING)

Currently pursuing Master's in Computing at The Open University.

Milton Keynes, England (Remote)

May 2024 - Ongoing

The King's University

B.S. IN COMPUTER SCIENCE

Edmonton, Alberta

Sep 2012 - Dec 2016

Skills

Backend Architecture & Multi-Tenant Systems	Design event-driven, multi-tenant architectures serving 5 teams with 99.9% uptime. Built systems handling millions of events with AWS (Lambda, EKS, SQS), microservices on Kubernetes, and APIs processing \$4.3M in transactions. Experience with domain-driven design and scalable backend patterns.
Technical Leadership & Domain Ownership	Own technical domains end-to-end: architecture, operations, SLOs, and on-call. Set standards for 12-person team through RFC authorship, code reviews, and mentorship. Member of Architecture Steering Committee. Lead projects from design through production with cross-functional collaboration.
Infrastructure & Platform Engineering	Built developer platforms (60x faster provisioning) and test frameworks (30% defect reduction). Infrastructure as code with Terraform. Production Kubernetes/EKS clusters with observability (Datadog, Elasticsearch). AWS multi-account architectures. Focus on reliability and operational excellence.
Databases & Async Processing	Experience with Postgres, Redis, and Elasticsearch. Built event-driven systems using SQS for async job processing. Designed data models, optimized performance, and ensured data consistency across distributed systems.

Technologies

Backend Languages	Python, Java, TypeScript, Node.js, Bash, C#
Architecture & APIs	Event-Driven Architecture, Multi-Tenant Systems, REST APIs, OpenAPI, Microservices
Databases	PostgreSQL, Aurora/RDS, Redis, Elasticsearch
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM), Terraform, Kubernetes, Docker, Helm
Async & Messaging	SQS, Lambda, Event-Driven Patterns
Observability	Datadog, Elasticsearch, Kibana, Metrics Design, Alerting